SAR-Conditioned Brownian-Bridge Reconstruction of All 13 Sentinel-2 Bands Under Clouds: A Controlled Study on SEN12MS-CR

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Abstract—Optical Sentinel-2 imagery is frequently obscured by clouds, while Sentinel-1 synthetic aperture radar (SAR) is not. We present dieWolke, a SAR-conditioned Brownian-bridge diffusion model that reconstructs all 13 Sentinel-2 bands, and test four proposed additions in a controlled setting: band-group attention over visible, red-edge/NIR and SWIR bands (C1), an NDVI-consistency loss (C2), a per-pixel Laplace uncertainty head (C3), and a separate cloud-mask stage instead of end-to-end training (C5). Every model is trained on 2,434 patches of the spring subset of SEN12MS-CR and evaluated on its full official spring test split (1,500 patches) with paired bootstrap confidence intervals. With single-step inference, our model reaches 28.12 dB PSNR on real clouds, 2.1 dB above a gradient-boosting gap-filling baseline and 0.5–0.7 dB below the published UnCRtainTS and GLF-CR checkpoints, which were trained on far more data. The separate mask stage is the clearest result: it beats end-to-end training by 1.09 dB PSNR and 0.078 SSIM, and masks from our Stage-1 masker are as good as exact ones. The uncertainty head ranks errors (Spearman 0.43–0.55), and keeping the 50 % most certain pixels lowers NDVI RMSE by 26 %. The NDVI loss lowers NDVI error by about 3.5 % at a 0.16 dB PSNR cost, and band-group attention does not help at this data scale. The deficit to the published models comes from partly cloudy patches; under near-total cloud all deep models are within 0.2 dB.

Index Terms—cloud removal, Sentinel-1, Sentinel-2, SAR–optical fusion, diffusion bridge, uncertainty, NDVI

I. Introduction

Sentinel-2 provides 13-band multispectral imagery every few days, but cloud cover makes many acquisitions unusable, most severely in the seasons and regions where crop monitoring matters most. Sentinel-1 SAR images the same ground through clouds. SAR–optical fusion networks such as DSen2-CR [2], GLF-CR [3] and UnCRtainTS [4] exploit this on the SEN12MS-CR benchmark [1], and recent generative models cast the task as a diffusion bridge from the cloudy to the clear image [5].

Downstream users of reconstructed imagery mostly consume vegetation indices and need to know which pixels to trust. This motivates the four additions we study on top of a SAR-conditioned Brownian bridge:

- C1, band-group attention: separate attention streams, stems and output heads for the visible, red-edge/NIR and SWIR bands;
- C2, NDVI-consistency loss: an L1 penalty on the NDVI implied by the reconstruction;
- C3, per-pixel uncertainty: a Laplace scale head trained on a detached mean and calibrated by temperature scaling;
- C5, separate mask stage: a Stage-1 cloud/shadow masker versus end-to-end training without any mask.

(C4, an Indian Kharif-season benchmark, is future work and is not evaluated here.)

Our contribution in this paper is a controlled evaluation of these components rather than a new state of the art. All of our own models share one architecture, one training budget and one data subset, so each ablation isolates one change. We report paired bootstrap confidence intervals for every difference, break the results down by cloud cover and by scene, and run the published UnCRtainTS and GLF-CR checkpoints through the same evaluation code. Negative results (C1) are reported as such.

II. Related Work

SAR–optical cloud removal. DSen2-CR [2] introduced a residual network that fuses Sentinel-1 with cloudy Sentinel-2 and the SEN12MS-CR dataset of co-registered S1, cloudy S2 and clear S2 triplets over four seasons [1]. GLF-CR [3] adds global–local SAR-guided fusion, and UnCRtainTS [4] combines attention with aleatoric uncertainty, both for mono- and multi-temporal input.

Generative cloud removal. DB-CR [5] models cloud removal as a diffusion bridge between the cloudy and the clear image with attention-based SAR fusion and reports single-step inference. Brownian-bridge diffusion for image-to-image translation was introduced by BBDM [6]. CloudBreaker [7] applies conditional flow matching from SAR alone to four optical bands and evaluates NDVI after reconstruction.

Classical gap filling. Gupta and Verma [8] reconstruct cloudy Sentinel-2 pixels with tree ensembles before crop

classification. Our gradient-boosting baseline [9] follows this spirit but is not their method.

III. Method

A. Data representation

Each sample is a 13-band cloudy image y, a clear target x_0 of the same location at a nearby date, and a two-channel Sentinel-1 image s (VV, VH). Sentinel-2 digital numbers are clipped to $[0, 10^4]$ and divided by 10^4; VV and VH are clipped to $[-25, 0]$ and $[-32.5, 0]$ dB and scaled to $[0, 1]$, following SEN12MS-CR conventions. A binary cloud/shadow mask m comes either from the data (Section III-G) or from the Stage-1 masker.

B. Brownian bridge

We use a Brownian bridge between x_0 at $t = 0$ and y at $t = 1$,

$$x_t = (1 - t)\, x_0 + t\, y + \sigma\sqrt{t(1 - t)}\,\epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (1)$$

with $\sigma = 0.5$. At $t = 1$ the state equals the cloudy image exactly, so sampling starts from y without noise. Given a prediction $\hat{x}_0$, the reverse step from t to $s < t$ has mean $\hat{x}_0 + \frac{s}{t}(x_t - \hat{x}_0)$ and variance $\sigma^2 s(t - s)/t$. With a single step, the sampler returns the network's prediction at $t = 1$; all results in this paper use this deterministic single-step regime. During training t is uniform on $[10^{-3}, 1]$, and with probability 0.25 it is set to exactly 1 so that the single-step case is trained directly.

C. Network

The denoiser is a U-Net with four resolution levels (48, 96, 192, 192 channels), two residual blocks per level and a sinusoidal time embedding. Its input is x_t, the optical condition (the cloudy image with masked pixels set to zero), the mask and s. A separate SAR encoder produces features at every level, and attention blocks at 1/4 and 1/8 resolution and in the bottleneck cross-attend from optical to SAR features. The output is a residual on the condition: clear pixels are copied and the network fills what is masked. The training loss is L1 on all 13 bands plus the C2 and C3 terms below.

D. C1: band-group attention

In the full model, the input stem, every attention block and the output head are split into three band groups: visible (B01–B04), red-edge/NIR (B05–B08, B8A, B09) and SWIR (B10–B12). Each group has its own projection, self-attention, SAR cross-attention and feed-forward layers; the streams exchange a pooled summary through a gate initialised to identity and are merged by a zero-initialised projection. Each group's bands are predicted by a group-specific head. The ablation no_bga replaces all of this by all-band equivalents with a similar parameter count.

E. C2: NDVI-consistency loss

We add $\lambda_{\text{ndvi}} \|\text{NDVI}(\hat{x}_0) - \text{NDVI}(x_0)\|_1$ with $\lambda_{\text{ndvi}} = 0.1$, where $\text{NDVI} = (\text{NIR} - \text{Red})/(\text{NIR} + \text{Red})$. The gradient of this ratio grows as the inverse of the denominator, and with a small epsilon it drove our first training runs to NaN. We therefore clamp the predicted denominator at 0.1 and exclude pixels whose target denominator is below 0.1 (water, shadow, no-data).

F. C3: per-pixel uncertainty

The head also outputs a per-band log-scale $\log b$, and the model is trained with the Laplace negative log-likelihood $|x_0 - \text{sg}(\hat{x}_0)|/b + \log b$ with weight 0.05, where sg stops the gradient [10]. Detaching the mean means the head only explains the remaining error and does not change what the mean learns. After training, one temperature per band is fitted on the validation split to maximise likelihood [11].

G. C5: cloud masks and the two-stage design

For real cloudy images the mask is estimated: s2cloudless [12] probability above 0.4, dilated, plus a spectral-index shadow heuristic [13] restricted to the neighbourhood of clouds. For synthetic training and evaluation pairs, a real cloud from another patch is blended onto a clear image with soft edges and a displaced cast shadow, which gives an exact mask. The Stage-1 masker is a four-level U-Net (base width 32) trained for 15k steps on synthetic clouds with binary cross-entropy plus 0.5 Dice loss; it reaches a validation IoU of 0.964. The end_to_end variant receives the raw cloudy image and no mask at all.

IV. Experimental Setup

A. Data

We use the spring subset of SEN12MS-CR (ROIs1158) with the official scene split. Training uses a seeded random sample of 2,500 patches of the training scenes; after skipping pairs whose "clear" target has more than 5 % cloud, 2,434 training pairs remain. Validation uses 300 patches and test 1,500 patches from the five official spring test scenes, all at 256×256 pixels. Spring-only training and testing is not comparable with results on all four seasons, such as Table I of DB-CR [5]; we therefore do not compare against numbers reported in other papers.

B. Training

All bridge models are trained with AdamW (learning rate 2×10^{-4}, weight decay 10^{-4}, 1,000 warm-up steps, cosine decay to 5 %), batch size 8, random 128×128 crops with flips and rotations, gradient clipping at 1.0, fp16 mixed precision and an exponential moving average (0.999) of the weights, on NVIDIA T4 GPUs. The full model and the matched baseline train for 40k steps; the ablations, a 20k-step copy of the full model (full_20k) and the end-to-end variant train for 20k steps. Each configuration is trained once (one seed). The checkpoint with the best validation PSNR is evaluated with its EMA weights.

C. Compared methods

Matched baseline: the same bridge model with C1–C3 removed. UnCRtainTS and GLF-CR: the authors' published checkpoints (UnCRtainTS monotemporalL2, trained with an L2 loss and without uncertainty output; GLF-CR CR_net), run unchanged through our evaluation. Both were trained on all seasons of SEN12MS-CR. As a check of our input conventions, UnCRtainTS scores 28.83 dB / 0.882 SSIM / 8.00° SAM on our spring test split, against 28.90 / 0.880 / 8.32 reported on the all-season test split [4]. XGBoost: per-patch gradient boosting from S1 and clear pixels to each S2 band [9], with a model pooled over the training split for the 837 test patches that have too few clear pixels.

D. Metrics

Following the SEN12MS-CR literature, PSNR, SSIM, MAE, RMSE and SAM (degrees) are computed on all 13 bands with data range 1, per image, then averaged. We add NDVI RMSE and versions of PSNR and NDVI RMSE restricted to masked (cloudy) pixels. Differences between methods are paired over test images, with 95 % bootstrap confidence intervals. Uncertainty is assessed by negative log-likelihood, calibration error, the Spearman correlation between predicted scale and absolute error, and the area under the sparsification error curve (AUSE) [14].

V. Results

A. Main comparison on real clouds

Table I lists the results on real clouds. All deep models improve PSNR by about 10 dB over the cloudy input and beat XGBoost by more than 2 dB. The full model is 0.20 dB [0.14, 0.27] below the matched baseline. The published UnCRtainTS and GLF-CR checkpoints are 0.71 dB [0.59, 0.81] and 0.46 dB [0.36, 0.57] above the full model; GLF-CR has the lowest NDVI error by a clear margin.

B. Where the gap comes from

Fig. 1 splits the test set by estimated cloud fraction. Under near-total cloud (fraction ≥ 0.9, 718 patches, 48 % of the test set), where only SAR carries information, the four deep models lie within 0.2 dB of each other: the full model is level with the matched baseline (+0.06 dB [−0.02, +0.14]) and GLF-CR (+0.12 [−0.02, +0.28]) and 0.20 dB [0.06, 0.33] behind UnCRtainTS. On partly cloudy patches (782 patches) the published models lead by 0.78–1.17 dB, which points at the use of visible pixels and the date difference between the cloudy and the clear acquisition, the parts that most likely benefit from more training data. The per-scene ranking also varies: on scene 44 both of our models beat UnCRtainTS, and on the cloudiest scene (140, mean cloud fraction 0.92) the full model beats both the matched baseline and GLF-CR. Because the spring test split contains only five scenes and patches within a scene are correlated, the per-image confidence intervals understate the uncertainty of the overall ranking.

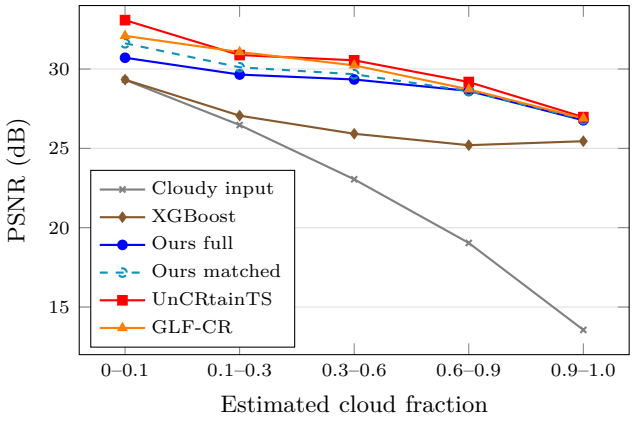


Fig. 1. PSNR on real clouds by estimated cloud fraction (bins of 139, 148, 196, 299 and 718 patches). The deep models converge under near-total cloud; the published models lead on partly cloudy patches.

C. Synthetic clouds

With transplanted clouds, ground truth and masks are exact (Table II). Here the full model is 0.20 dB [0.12, 0.28] above the matched baseline, with lower SAM and NDVI RMSE, while the matched baseline keeps a higher SSIM. The sign of the full-versus-matched difference therefore depends on the cloud type, and neither model is consistently better.

D. Ablations of C1–C3

Table III removes one component at a time from full_20k. Every removal raises whole-image PSNR: by 0.36 dB [0.29, 0.44] without band-group attention, 0.16 [0.11, 0.20] without the NDVI loss and 0.15 [0.09, 0.21] without the uncertainty head. The NDVI loss is the only component that lowers its target metric: without it, whole-image NDVI RMSE rises by 0.0053 [0.0038, 0.0067], about 3.5 %; inside clouds the effect is not significant. Band-group attention does not lower the error of the red-edge, NIR or SWIR bands it was designed for: per-band MAE of no_bga is equal or lower in B05–B12 (e.g. B07 0.0563 vs. 0.0587, B8A 0.0595 vs. 0.0619).

E. C5: separate mask stage versus end-to-end

Table IV compares, at matched training length, the model given exact masks, the same model given the Stage-1 masker's masks, and the end-to-end variant without any mask. End-to-end training loses 1.09 dB PSNR [0.98, 1.20], 0.078 SSIM [0.076, 0.081] and 1.36 dB inside clouds [1.24, 1.48]; it is better on only 26 % of images by PSNR and 3 % by SSIM. Its error in B01 (coastal aerosol) is almost twice that of the masked model. Its NDVI RMSE is slightly lower (−0.0049 [−0.0081, −0.0016]). Masks from the Stage-1 masker match exact masks at both training lengths, so the two-stage design costs nothing in accuracy.

TABLE I
Real clouds, SEN12MS-CR spring test split (1,500 patches). Mean over images; "cloud" columns use masked pixels only. Our models are trained on 2,434 spring patches; UnCRtainTS and GLF-CR use the authors' checkpoints trained on all seasons.

Method	PSNR↑	SSIM↑	MAE↓	RMSE↓	SAM↓	NDVI RMSE↓	PSNR cloud↑	NDVI RMSE cloud↓
Cloudy input	18.624	0.6761	0.1247	0.1616	13.817	0.2686	17.730	0.2867
XGBoost	25.981	0.8028	0.0384	0.0532	9.535	0.1669	25.699	0.1765
Ours, full (C1+C2+C3)	28.124	0.8664	0.0292	0.0431	8.403	0.1530	28.033	0.1503
Ours, matched baseline	28.324	0.8699	0.0286	0.0423	8.319	0.1522	28.188	0.1500
UnCRtainTS (published)	28.830	0.8820	0.0280	0.0400	7.999	0.1449	28.511	0.1468
GLF-CR (published)	28.588	0.8867	0.0279	0.0406	7.789	0.1286	28.600	0.1292

TABLE II
Synthetic clouds (exact masks), test split.

Method	PSNR	SSIM	SAM	NDVI RMSE	PSNR cloud
Cloudy input	19.510	0.6887	8.808	0.2258	17.208
Ours, full	29.080	0.8699	7.491	0.1478	27.926
Ours, matched	28.880	0.8761	7.616	0.1509	27.636

TABLE III
Ablations on real clouds, 20k training steps each. Δ is the paired difference to full_20k with 95 % CI.

Run	PSNR	SAM	NDVI RMSE	ΔPSNR
full_20k	27.839	8.690	0.1518	—
no_bga (−C1)	28.203	8.653	0.1529	+0.36 [0.29, 0.44]
no_ndvi (−C2)	27.997	8.624	0.1571	+0.16 [0.11, 0.20]
no_unc (−C3)	27.986	8.426	0.1494	+0.15 [0.09, 0.21]

TABLE IV
C5 on synthetic clouds. All rows except the last two use 20k training steps.

Configuration	PSNR	SSIM	SAM	PSNR cloud
full_20k, exact masks	28.746	0.8625	7.952	27.556
full_20k, Stage-1 masks	28.757	0.8642	7.948	27.582
end_to_end, no mask	27.658	0.7841	9.145	26.196
full (40k), exact masks	29.080	0.8699	7.491	27.926
full (40k), Stage-1 masks	29.084	0.8713	7.485	27.947

TABLE V
Uncertainty of the full model on real clouds, and NDVI RMSE on the pixels kept after removing the most uncertain ones.

	NLL	Calib. err.	Spearman	AUSE
Raw	−2.209	0.085	0.431	0.00606
Temperature-calibrated	−2.230	0.055	0.440	0.00593
Pixels kept	100 %	90 %	75 %	50 %
NDVI RMSE	0.182	0.171	0.158	0.134

F. C3: uncertainty

On real clouds the predicted scale correlates with the absolute error (Spearman 0.431, 0.440 after calibration), and temperature scaling lowers the calibration error from 0.085 to 0.055 (Table V). Dropping the most uncertain pixels steadily lowers NDVI error, from 0.182 at full coverage to 0.134 when the 50 % most certain pixels are kept, a 26 % reduction. On synthetic clouds the raw calibration error is 0.037 and the Spearman correlation 0.55, but the temperature fitted on real-cloud validation data worsens calibration to 0.060: calibration does not transfer between cloud types. Neither the matched baseline nor the UnCRtainTS checkpoint used here outputs uncertainty, so no uncertainty comparison is possible.

VI. Discussion and Limitations

Training data. Our models see 2,434 spring patches, while the published checkpoints were trained on the full four-season SEN12MS-CR training split. Their lead is concentrated on partly cloudy patches, which require learning how to combine visible pixels with SAR across a date gap. Whether this gap is due to data or to architecture cannot be decided from these runs; retraining on a larger part of the spring training split, which has 24,378 patches in total, is under way.

Band-group attention. At this data scale C1 lowers PSNR and does not help the bands it targets. The additional per-group parameters may need more data, or the grouping may simply not be a useful inductive bias for this task; we do not claim C1 as a contribution on the basis of these results.

Statistical caveats. Each configuration is trained with a single seed, so the confidence intervals cover variation over test images, not over training runs; differences of a few tenths of a dB between our own models may not survive another seed. The spring test split covers only five scenes. Real-cloud masks are estimates (s2cloudless plus a shadow heuristic), and the validation split ranked the ablations differently from the test split.

Comparability. All numbers are for the spring subset and are not comparable with all-season results in the literature, including DB-CR [5].

VII. Conclusion

In a controlled single-step setting on SEN12MS-CR spring, a SAR-conditioned Brownian bridge reconstructs all 13 Sentinel-2 bands to within 0.5–0.7 dB of published models trained on far more data and more than 2 dB above classical gap filling. A separate cloud-mask stage is clearly better than end-to-end training, and a learned Stage-1 masker is as good as exact masks. A detached Laplace

uncertainty head gives a usable reliability signal that substantially lowers NDVI error when uncertain pixels are filtered. The NDVI-consistency loss trades a small PSNR loss for a small NDVI gain, and band-group attention is not supported at this scale. Next steps are training on more of the spring training split, repeating key runs with additional seeds, and building an Indian Kharif-season benchmark with downstream crop classification.

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